

Group Retrospective Summary



Dataspaces Project Team

Team **Pink Panther**

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Version Control

Version	Date	Author	Amendments	Status
1	06.03	Pink Panther	Initial creation	Done & Submitted
2	23.03	Pink Panther	Sprint 1	Completed
3	13.04	Pink Panther	Sprint 2	Polished
4	11.05	Pink Panther	Sprint 3	Incerted
5	08.06	Pink Panther	Sprint 4	Draft
6	11.06	Pink Panther	Styling	Solid

Sprint 0

This sprint has been highly effective in establishing a strong team foundation.

Key accomplishments include:

- **Team Dynamics:** Identifying individual strengths and weaknesses, and establishing clear, concrete roles.
- **Workload Management:** Ensuring a balanced and equitable workload distribution across the team.
- **Communication & Collaboration:** Maintaining excellent standards for document sharing and overall communication.

We are committed to sustaining this level of collaboration and positive performance moving forward.

Sprint 1

What went well

This sprint was highly effective in establishing a strong team foundation.

- Team dynamics:
The team successfully identified individual strengths and weaknesses and established clear, concrete roles.
- Workload management:
Tasks were distributed in a balanced and equitable manner, ensuring steady progress across all areas.
- Communication & collaboration:
Strong communication was maintained through regular meetings and consistent document sharing. The team remained well-aligned throughout the sprint.
- Engagement with coaches and stakeholders:
Frequent interactions helped improve understanding of expectations and project constraints.

What we did

During this sprint, the team held multiple meetings with stakeholders, coaches, and internally to better understand the project direction and expectations.

- A meeting with coaches and Martijn focused on assessing and evaluating whether the current project direction supports achieving team KPIs.
- Across all meetings, the team worked on clarifying:
 - What value can realistically be delivered
 - What stakeholders expect from the project

Challenges

- Unclear project direction:
The team experienced uncertainty regarding scope, expectations, and priorities.
- Dual-technology requirement (FIWARE + EDC):
The requirement to implement and compare two systems for the same purpose significantly increases:
 - Complexity
 - Workload
 - Risk of shallow technical depth
- This is particularly challenging since:
 - Both systems are already partially implemented
- Technical uncertainty:
Gaps in documentation and unfamiliar technologies made it difficult to define a clear implementation path.
- KPI alignment concerns:
The current scope may limit the team's ability to meet individual learning goals.

Key discussions & decisions

- Based on feedback from coach Frank:
 - The team will create a pros and cons comparison document for FIWARE and EDC
 - This will be used to support a structured discussion with stakeholders
- During the session with coaches and Martijn, three possible paths were discussed:
 - Continue as planned, but shift focus toward individual deliverables
 - Advocate for focusing solely on EDC
 - Consider reassignment to a different project
- Agreed approach:

The team will make one final attempt to align with stakeholders on a workable solution.

 - If alignment is achieved -> continue with the project
 - If not -> consider switching projects

What we learned

- Early and clear alignment with stakeholders is essential
- A focused scope is necessary for both project success and individual learning
- Technical clarity significantly improves efficiency
- Regular communication with coaches helps navigate uncertainty

Action points for next sprint

- Finish the use cases for FIWARE
- EDC initial setup & test cases
- Align project scope with team KPIs and individual goals
- Define clearer:
 - Deliverables
 - Use cases
 - Technical approach

Overall reflection

Despite uncertainties in project direction, the team demonstrated strong collaboration, adaptability, and communication. The focus for the next sprint is to achieve clarity and ensure the project setup supports both team success and individual development.

Sprint 2

What went well

This sprint marked a significant improvement in both technical clarity and team alignment compared to previous iterations.

- **Clearer architectural understanding:**

The team successfully moved from a tool-focused approach (EDC vs FIWARE) to a broader understanding of the dataspace ecosystem, including the role of additional components such as identity and control layers.

- **Successful documentation output:**

Multiple key documents were completed, including:

- Eclipse Dataspace Selection & Gaia-X Alignment
- FIWARE Technical Documentation (Research v5)

These documents helped consolidate knowledge and create a shared technical reference point.

- **Improved task tracking (Trello):**

The introduction of a structured Trello board improved visibility of:

- completed work
- current priorities
- blockers

This reduced ambiguity in responsibilities and progress tracking.

- **Better division of responsibilities:**

Team members began focusing more clearly on their domains (e.g., IAM, EDC, documentation), leading to more efficient parallel work.

What we did

During this sprint, the team focused on consolidating research, validating previous work, and improving project structure.

- Finalized the **EDC vs FIWARE architectural decision**, formally documenting the reasoning behind selecting EDC as the primary framework.
- Produced a detailed **technical breakdown of the FIWARE Dataspace Connector**, including identity flows, marketplace orchestration, and deployment challenges.
- Created and structured a **Trello board** to track all project activities, including completed, ongoing, and blocked tasks.
- Refined understanding of **Gaia-X alignment**, including decentralized identity (DID/VC) and policy-based governance.
- Identified that a complete dataspace architecture extends beyond EDC, introducing awareness of additional components such as identity and orchestration layers.

Challenges

Despite progress, several challenges remained:

- **Architectural ambiguity from stakeholders:**

There is still uncertainty regarding whether the expected implementation should focus solely on EDC or include additional Eclipse Dataspace components (e.g., identity/control plane).

- **Incomplete identity (IAM) implementation:**

Current setups rely on partial or mock identity solutions, while full DID/VC-based flows remain complex and not fully implemented.

- **FIWARE complexity and limitations:**

While FIWARE components were successfully deployed and analyzed, full end-to-end flows required significant manual configuration and revealed architectural mismatches with Gaia-X goals.

- **Transition from research to implementation:**

The team has strong theoretical understanding, but translating this into a fully integrated, production-like dataspace remains a challenge.

Key discussions & decisions

Decision to focus on EDC as the core framework:

Based on research and architectural comparison, the team confirmed EDC as the primary technology moving forward.

Recognition of broader ecosystem (important shift):

The team identified that:

- EDC is the **core connector**
- Additional components (identity, orchestration) are required for a complete dataspace

Shift from “building extensions” to “integrating components”:

A key discussion point was whether to:

- develop custom extensions
- or integrate existing Eclipse Dataspace projects

This remains an open architectural consideration to validate with stakeholders.

Adoption of structured task management (Trello):

The team agreed to formalize task tracking to avoid previous issues with unclear responsibilities and progress visibility.

What we learned

- **A dataspace is an ecosystem, not a single tool**
- Identity (DID/VC) is one of the most complex and critical parts of the architecture
- Clear architectural understanding significantly reduces wasted effort
- Documentation is essential for aligning both internally and with stakeholders
- Early assumptions about scope can lead to unnecessary complexity if not validated

Action points for next sprint

Clarify architectural expectations with stakeholders

- Confirm whether integration with additional Eclipse Dataspace components is required

Start implementing real IAM (DID/VC)

- Replace mock identity setup with a more realistic solution

Move from research to implementation

- Focus on achieving at least one fully working end-to-end dataspace flow

Continue improving Trello usage

- Keep tracking blockers and completed work consistently

Define final use case(s) clearly

- Ensure alignment between technical implementation and project goals

Overall reflection

This sprint represents a transition from **exploration to structured execution**.

While previous sprints focused on understanding tools and resolving uncertainty, this sprint established:

- a clear technical direction (EDC-centered architecture)
- improved project organization (Trello + documentation)
- a deeper understanding of dataspace architecture

The next step is to **translate this clarity into a working, integrated system**, with a particular focus on identity.

Sprint 3

What went well

This sprint marked the team's transition from architectural planning to practical implementation and deployment activities.

- Successful Eclipse MVD deployment
- The team successfully deployed the Eclipse Minimum Viable Dataspace (MVD) environment locally and gained hands-on experience with the Eclipse Dataspace ecosystem.
- Successful FIWARE MVD deployment
- The team successfully deployed the FIWARE Minimum Viable Dataspace (MVD) environment in the Fontys netlab and gained hands-on experience with the FIWARE Dataspace ecosystem.
- Improved understanding of deployment infrastructure
- Working with Docker, Kubernetes, Squid-proxy, Keycloak, KinD, PostgreSQL, Identity Hub, and related components significantly improved the team's understanding of how a dataspace is deployed and operated within a containerized environment.
- Strong collaboration during troubleshooting
- Several deployment and networking challenges emerged during the sprint. Team members collaborated effectively to investigate issues, share findings, and test alternative approaches.

- Productive stakeholder and coach feedback
- Documentation reviews and technical discussions with coaches and stakeholders provided valuable feedback that helped improve both technical accuracy and documentation quality.

What we did

During this sprint, the team focused on deploying, validating, and documenting both the Eclipse and FIWARE MVD environment.

Deployment activities

- Installed and configured the Eclipse/FIWARE MVD environment.
- Worked with Kubernetes-based deployments using KinD for eclipse and Maven for FIWARE.
- Investigated deployment on remote VM environments (netlab).

Testing and validation

- Executed end-to-end dataspace flows using the Bruno collections (eclipse).
- Executed and tested end-to-end dataspace flows using the Local.md file located in the dataspace-connector github repository from FIWARE.
- Verified asset registration and discovery.
- Tested contract negotiation workflows.
- Validated successful data transfer processes.

Documentation

The team produced multiple technical deliverables, including:

- Eclipse & FIWARE MVD Deployment
- Updated Dataspace documentation
- Eclipse & FIWARE Testing Report
- Supporting architecture and implementation notes

Infrastructure investigation

- Investigated VM deployment limitations. (CPU, RAM and storage)
- Analyzed networking, DID resolution and JWT decoding issues.
- Explored potential production deployment requirements.

Challenges

- Networking complexity
- Deploying distributed dataspace components introduced networking challenges, particularly when attempting to expose services beyond the local environment.

With the particular case for FIWARE being the netlab vm resource restrictions. We had to contact the network admin to resolve this issue, which took quite some time.
- DID resolution limitations
- The team discovered that decentralized identity resolution becomes significantly more complex when moving beyond local deployments.
- JWT decode error
- The FIWARE team collaborated well on a very specific error which was an endpoint not being able to decode a JWT token.

This took quite some time to figure out how to fix. But was eventually fixed.
- Contacting FIWARE developers
- We tried contacting the FIWARE development team like we did before. Unfortunately they took several weeks to respond, having us spend a lot of time looking around when perhaps the solution for the decode problem could be given to us by them.
- Infrastructure uncertainty
- Several deployment approaches were explored for eclipse before identifying a stable and repeatable local setup.
- Balancing implementation and documentation
- Considerable effort was required to both implement the system and maintain high-quality documentation simultaneously.

Key discussions & decisions

Focus on validating the complete flow

Rather than implementing custom features immediately, the team focused on proving that the entire dataspace workflow operated correctly.

Documentation quality improvements

Following stakeholder feedback, the team agreed to:

- Improve evidence collection.
- Add stronger validation examples.
- Increase consistency across project documentation.
- Improve terminology accuracy.

What we learned

- Deployment is a major part of dataspace projects
- Successfully deploying and operating the environment required substantially more effort than initially anticipated.
- Identity remains one of the most important challenges
- DID and trust management continue to be critical aspects of dataspace architecture.
- Correctly structuring JWT tokens in a VC context within FIWARE
- We learned how the token is supposed to be structured so that the backend system can gracefully decode it properly for the 'credential handling'.

- Working systems provide better insight than theoretical research
- Many architectural concepts became significantly easier to understand once the team interacted with a real implementation.
- Evidence is essential
- Technical claims are stronger when supported by screenshots, logs, demonstrations, and reproducible validation steps.

Action points for next sprint

- Complete remaining deployment validation activities.
- Improve documentation based on received feedback.
- Develop a dashboard concept for visualizing dataspace interactions.
- Investigate user experience improvements.
- Demonstrate the complete dataspace workflow visually.
- Prepare final project deliverables.

Overall reflection

This sprint represented the transition from architectural understanding to practical implementation. The team successfully deployed and validated a functioning dataspace environment for both eclipse and FIWARE while gaining valuable experience with real-world deployment challenges. Although infrastructure and networking issues created obstacles, the sprint significantly improved the team's technical maturity and understanding of dataspace ecosystems.

Sprint 4

What went well

This sprint focused on finalizing project deliverables, improving documentation quality, and transforming technical findings into a presentable solution.

- Strong improvement cycle
- The team responded effectively to stakeholder and reviewer feedback, resulting in significant improvements across multiple project documents.
- Increased project maturity
- By this stage, the team had developed a clear understanding of the technical aspects of dataspace and have contributed with their own developments.
- Better presentation of results
- The team shifted focus from describing work performed to demonstrating evidence that solutions functioned as intended.
- Successful collaboration
- Responsibilities were clearly divided, enabling parallel progress on implementation, documentation, and presentation materials.

What we did

Documentation refinement

The team reviewed and improved:

- Deployment & Testing Report
- Eclipse Dataspace Selection & Gaia-X Alignment
- Supporting technical documents

Dashboard planning

The team explored the concept of a dataspace dashboard capable of visualizing:

- Catalog requests
- Contract negotiations
- Transfer processes
- Data access operations
- Showing corresponding DSSC building blocks

Final validation

- Reviewed deployment results.
- Verified technical findings.
- Cross-checked conclusions with project goals and stakeholder expectations.
- Stakeholder feedback

Presentation preparation

- Prepared project poster materials.
- Organized project documentation.
- Consolidated lessons learned and future recommendations.

Challenges

- Demonstrating complex concepts to non-technical audiences
- Dataspace involve advanced concepts such as identity, trust, policy enforcement, and sovereignty that are difficult to communicate clearly to a lesser technical audience.
- Finalizing documentation
- Maintaining consistency across multiple documents required significant review effort.
- Scope management
- The team continuously balanced technical ambition against available project time.
- Translating technical success into stakeholder value
- Demonstrating why the implementation matters was often more challenging than implementing the technology itself.

Key discussions & decisions

Focus on visualization and transparency

The team identified that future dataspace solutions would benefit from improved monitoring and visualization capabilities.

Emphasis on evidence-based reporting

Documentation should demonstrate outcomes rather than simply describe activities.

Recognition of future opportunities

The team discussed potential extensions including:

- Advanced IAM integration - Educloud
- Enhanced monitoring
- Production-grade deployments
- Dashboard-based observability

What we learned

- Dataspaces combine technology and governance
- Success depends not only on technical implementation but also on trust, policy, and organizational alignment.
- Clear communication is critical
- Complex systems become significantly more valuable when their behavior can be easily understood by stakeholders.
- Documentation is part of engineering
- Producing accurate, evidence-based documentation is as important as implementing the solution itself.
- Continuous feedback improves outcomes
- Regular stakeholder reviews significantly improved both the technical and presentation quality of the project.

Action points for project closure

- Finalize all deliverables.
- Submit final documentation.
- Demonstrate deployment results.
- Document future recommendations.
- Archive project knowledge for future teams.

Overall reflection

Sprint 4 focused on consolidating the team's work and preparing the project for final delivery. The sprint demonstrated how the project evolved from initial uncertainty into a structured and validated implementation of dataspace technologies. Beyond the technical achievements, the team developed a deeper understanding of collaboration, stakeholder management, documentation quality, and the broader role of dataspace in secure digital ecosystems.